B5 General Relativity & Cosmology — Model Solutions, 2016

Question 1 — Schwarzschild orbits and circular geodesics

Problem. Schwarzschild metric. Reduce to equatorial plane; find $\dot t$, $\dot\phi$ and the orbit equation $u''+u = A+Bu^2$. A satellite is launched azimuthally from a stationary station at $r = R$ with locally measured speed $v_{\rm circ}$ giving a circular orbit. Find $v_{\rm circ}$ and the time before the orbiting satellite collides with the station. Sketch the perturbed orbit when the launch speed is $v_{\rm circ} + \Delta v$.

(a) Equatorial reduction and conserved quantities

The metric is spherically symmetric, so it is invariant under $\theta \to \pi - \theta$. Pick initial conditions with $\theta = \pi/2$, $\dot\theta = 0$ (always possible by rotation): the Euler–Lagrange equation for θ has the right-hand side proportional to $\sin\theta\cos\theta\,\dot\phi^2$, vanishing on the equator. Hence $\theta(\tau) \equiv \pi/2$ is a geodesic; restricting to it loses no generality.

Lagrangian per unit rest mass on the equator (using $\dot{} = {\rm d}/{\rm d}\tau$):

$$\mathcal{L} = -\big(1 - \tfrac{2GM}{c^2r}\big)c^2\dot t^{\,2} + \big(1 - \tfrac{2GM}{c^2r}\big)^{-1}\dot r^{\,2} + r^2\dot\phi^{\,2}. \tag{1}$$

$\mathcal{L}$ has no explicit t or ϕ:

$$\frac{\partial\mathcal{L}}{\partial\dot t} = -2c^2\big(1 - \tfrac{2GM}{c^2r}\big)\dot t = \text{const},$$

$$\frac{\partial\mathcal{L}}{\partial\dot\phi} = 2r^2\dot\phi = \text{const}.$$

Define the two constants of motion (energy E and angular momentum ℓ per unit rest mass):

$$\boxed{\big(1 - \tfrac{2GM}{c^2r}\big)\dot t = \frac{E}{c^2}, \qquad r^2\dot\phi = \ell.} \tag{2}$$

(b) Radial equation and orbit equation

Normalisation of the 4-velocity, $g_{ab}\dot x^a\dot x^b = -c^2$, on the equator:

$$-\big(1 - \tfrac{2GM}{c^2r}\big)c^2\dot t^{\,2} + \big(1 - \tfrac{2GM}{c^2r}\big)^{-1}\dot r^{\,2} + r^2\dot\phi^{\,2} = -c^2.$$

Substitute (2):

$$-\frac{E^2/c^2}{1 - 2GM/(c^2r)} + \frac{\dot r^{\,2}}{1 - 2GM/(c^2r)} + \frac{\ell^2}{r^2} = -c^2.$$

Multiply by $1 - 2GM/(c^2r)$ and rearrange:

$$\boxed{\dot r^{\,2} = \frac{E^2}{c^2} - \big(1 - \tfrac{2GM}{c^2r}\big)\left(c^2 + \frac{\ell^2}{r^2}\right).} \tag{3}$$

Change variable to $u(\phi) = 1/r$. Use $\dot r = (\mathrm{d}r/\mathrm{d}\phi)\dot\phi = -u^{-2}u'\,\ell u^2 = -\ell u'$:

$$\dot r^{\,2} = \ell^2 (u')^2.$$

Insert into (3):

$$\ell^2 (u')^2 = \frac{E^2}{c^2} - c^2 + \tfrac{2GM}{c^2} c^2 u - \ell^2 u^2 + \tfrac{2GM}{c^2}\ell^2 u^3.$$

Differentiate w.r.t. ϕ and divide by $2u'$:

$$\ell^2 u'' = GM - \ell^2 u + 3GMu^2.$$

Divide by ℓ^2:

$$\boxed{u'' + u = A + Bu^2, \qquad A = \frac{GM}{\ell^2}, \quad B = \frac{3GM}{c^2}.} \tag{4}$$

(c) Circular orbit launched from the station

Circular orbit: $u = u_0 = 1/R$ constant, so (4) gives $u_0 = A + Bu_0^2$:

$$\frac{1}{R} = \frac{GM}{\ell^2} + \frac{3GM}{c^2 R^2}.$$

Solve for ℓ^2:

$$\ell^2 = \frac{GM\,R^2}{R - 3GM/c^2} = \frac{GMR^2}{R(1 - 3GM/(c^2R))}.$$

Cleaner form:

$$\ell^2 = \frac{GMR}{1 - 3GM/(c^2R)}. \tag{5}$$

The astronaut's locally measured speed is the proper distance moved per unit proper time. The station is static at $r = R$, so the astronaut's frame uses

$$\mathrm{d}\ell_\perp = R\,\mathrm{d}\phi, \qquad \mathrm{d}\tau_{\mathrm{stn}} = \sqrt{1 - \tfrac{2GM}{c^2R}}\,\mathrm{d}t.$$

The satellite passes the station with $\dot r = 0$, $\dot\theta = 0$, $\dot\phi \neq 0$. Local speed:

$$v_{\mathrm{circ}} = \frac{R\,\mathrm{d}\phi}{\mathrm{d}\tau_{\mathrm{stn}}} = \frac{R\,\dot\phi}{\sqrt{1 - 2GM/(c^2R)}\,\dot t}.$$

From (2), $\dot\phi/\dot t = \ell c^2/(R^2E)\cdot(1 - 2GM/(c^2R))$. Use the alternative route: write $v_{\mathrm{circ}}^2 = (R\,\mathrm{d}\phi/\mathrm{d}\tau_{\mathrm{stn}})^2$. We have

$$\frac{(R\,\mathrm{d}\phi)^2}{\mathrm{d}\tau_{\mathrm{stn}}^2} = \frac{R^2\dot\phi^2}{(1 - 2GM/(c^2R))\,\dot t^{\,2}}.$$

Multiply top and bottom by $\mathrm{d}\tau^2$ and use $r^2\dot\phi = \ell$:

$$v_{\mathrm{circ}}^2 = \frac{\ell^2/R^2}{(1 - 2GM/(c^2R))\,\dot t^{\,2}/\mathrm{d}\tau^2}\cdot\frac{1}{R^2/R^2}.$$

Cleaner: along the circular orbit, the 4-velocity normalisation reads

$$-c^2 = -\big(1 - \tfrac{2GM}{c^2R}\big)c^2\dot t^{\,2} + R^2\dot\phi^{\,2}.$$

Solve for $R^2\dot\phi^2$:

$$R^2\dot\phi^2 = \left(1 - \tfrac{2GM}{c^2 R}\right)c^2\dot t^2 - c^2.$$

Divide by $(1 - 2GM/(c^2R))\,\dot t^2$:

$$v_{\rm circ}^2 = \frac{R^2\dot\phi^2}{(1 - 2GM/(c^2R))\,\dot t^2} = c^2 - \frac{c^2}{(1 - 2GM/(c^2R))\,\dot t^2}.$$

Use (2): $(1 - 2GM/(c^2R))\,\dot t = E/c^2$, so $(1 - 2GM/(c^2R))\,\dot t^2 = E^2/[c^4(1 - 2GM/(c^2R))]$. Cleaner: rewrite using $\ell = R^2\dot\phi$:

$$v_{\rm circ}^2 = \frac{\ell^2/R^2}{(1 - 2GM/(c^2R))\,\dot t^2}.$$

Expand $\dot t^2$ from the same normalisation: $\dot t^2 = (1 + \ell^2/(c^2R^2))/(1 - 2GM/(c^2R))$. Insert:

$$v_{\rm circ}^2 = \frac{\ell^2/R^2}{(1 - 2GM/(c^2R))\cdot(1 + \ell^2/(c^2R^2))/(1 - 2GM/(c^2R))} = \frac{\ell^2/R^2}{1 + \ell^2/(c^2R^2)}.$$

Substitute (5):

$$\frac{\ell^2}{R^2} = \frac{GM/R}{1 - 3GM/(c^2R)}.$$

Numerator divided by $1 + \ell^2/(c^2R^2)$. Compute $\ell^2/(c^2R^2) = (GM/(c^2R))/(1 - 3GM/(c^2R))$. Hence

$$1 + \frac{\ell^2}{c^2R^2} = \frac{1 - 3GM/(c^2R) + GM/(c^2R)}{1 - 3GM/(c^2R)} = \frac{1 - 2GM/(c^2R)}{1 - 3GM/(c^2R)}.$$

Divide:

$$v_{\rm circ}^2 = \frac{GM/R}{1 - 3GM/(c^2R)}\cdot\frac{1 - 3GM/(c^2R)}{1 - 2GM/(c^2R)}.$$

Simplify:

$$\boxed{v_{\rm circ}^2 = \frac{GM/R}{1 - 2GM/(c^2R)}.} \tag{6}$$

Note the Newtonian limit $v_{\rm circ}^2 \to GM/R$ when $2GM/(c^2R) \ll 1$.

Time to collision (in the astronaut's proper time). Orbital period in coordinate time: $T_t = 2\pi/\dot\phi\cdot(\dot t/1)$, but easier to relate proper times directly. Proper-time period of the satellite over one revolution:

$$T_{\rm sat} = 2\pi/\dot\phi = 2\pi R^2/\ell.$$

Substitute (5):

$$T_{\rm sat} = 2\pi R^2/\sqrt{GMR/(1 - 3GM/(c^2R))} = 2\pi\sqrt{R^3/GM}\,\sqrt{1 - 3GM/(c^2R)}.$$

Convert to station proper time via $\mathrm{d}\tau_{\rm stn} = \sqrt{1 - 2GM/(c^2R)}\,\mathrm{d}t$ and $\mathrm{d}t/\mathrm{d}\tau_{\rm sat} = \dot t$. Use (2):

$$\dot t = \frac{E/c^2}{1 - 2GM/(c^2R)}.$$

Energy from $\dot t^2 = (1 + \ell^2/(c^2R^2))/(1 - 2GM/(c^2R))$ and $E = c^2(1 - 2GM/(c^2R))\dot t$:

$$E^2/c^4 = (1 - 2GM/(c^2R))(1 + \ell^2/(c^2R^2)) = \frac{(1 - 2GM/(c^2R))^2}{1 - 3GM/(c^2R)}.$$

Coordinate-time period: $T_t = T_{\rm sat}\,\dot t = T_{\rm sat}\cdot E/[c^2(1-2GM/(c^2R))]$. Combine:

$$T_t = 2\pi\sqrt{R^3/GM}\,\sqrt{1-3GM/(c^2R)}\cdot\frac{E/c^2}{1-2GM/(c^2R)}.$$

Insert $E/c^2 = (1-2GM/(c^2R))/\sqrt{1-3GM/(c^2R)}$:

$$T_t = 2\pi\sqrt{R^3/GM}.$$

This is exactly Kepler's third law in coordinate time. Now convert to station proper time:

$$T_{\rm stn} = T_t\sqrt{1-2GM/(c^2R)}.$$

$$\boxed{T_{\rm stn} = 2\pi\sqrt{\frac{R^3}{GM}}\,\sqrt{1-\frac{2GM}{c^2R}}.} \tag{7}$$

Newtonian limit: $T_{\rm stn}\to 2\pi\sqrt{R^3/GM}$. Strong-field correction: time runs slower at the station than at infinity, so the wait is shorter than Kepler's value.

(d) Perturbed launch

For $|\Delta v| \ll v_{\rm circ}$ the orbit is a near-circular ellipse: writing $u = u_0+\delta u$ in (4) and linearising,

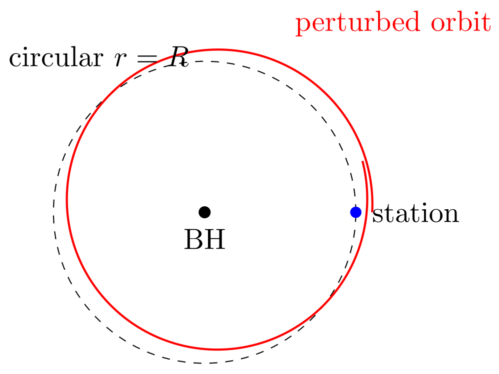
$$\delta u'' + (1-2Bu_0)\,\delta u = 0.$$

Substitute $Bu_0 = 3GM/(c^2R)$:

$$\delta u'' + \left(1-\tfrac{6GM}{c^2R}\right)\delta u = 0.$$

Apsidal precession frequency (per orbit) is $\sqrt{1-6GM/(c^2R)}<1$, so radial oscillations close in azimuthal angle $2\pi/\sqrt{1-6GM/(c^2R)}>2\pi$: each radial period the perihelion advances by

$$\Delta\phi_{\rm prec}\approx \pi\,\frac{6GM}{c^2R}.$$



The orbit oscillates between $r_- < R < r_+$, with the line of apsides precessing forward; for $\Delta v \neq 0$ the satellite returns to $r = R$ but at a shifted azimuth, so it does *not* hit the station on the first pass. However the perihelion drifts secularly, and after $\sim 2\pi/\Delta\phi_{\rm prec}$ radial periods the apse returns to the station's azimuth: the station is therefore *not* safe forever — collision occurs after a finite (but long) time. (Strictly true outside the ISCO at $r = 6GM/c^2$; below this the perturbation is unstable and collision is rapid.)

Question 2 — Einstein equations and TOV; uniform-density star

Problem. Argue $G_{ab} = R_{ab} - \frac{1}{2}g_{ab}R$. For a static spherical metric reduce $G_{ab} = -(8\pi G/c^4)T_{ab}$ to the TOV equations. For a uniform-density star derive $p_0(R)$ and the Buchdahl bound $2GM/(c^2R) < 8/9$. Comment on a redshift $z = 3.1$ neutron star.

(a) Form of the Einstein tensor

Energy–momentum conservation in curved space requires $\nabla_a T^{ab} = 0$. If $G^{ab} \propto T^{ab}$ then $\nabla_a G^{ab} = 0$.

The Ricci tensor alone fails: the contracted Bianchi identity gives $\nabla_a R^{ab} = \frac{1}{2}\nabla^b R \neq 0$. Subtract the trace to enforce conservation:

$$G_{ab} = R_{ab} - \tfrac{1}{2}g_{ab}R.$$

Bianchi identity then yields $\nabla^a G_{ab} \equiv 0$. G_{ab} is symmetric (both R_{ab} and g_{ab} are) and built from the metric and its first two derivatives, matching the second-derivative structure of T_{ab} (Newtonian limit: Poisson). The proportionality constant is fixed by Newtonian limit to $-8\pi G/c^4$:

$$\boxed{G_{ab} = R_{ab} - \tfrac{1}{2}g_{ab}R = -\tfrac{8\pi G}{c^4}T_{ab}.}$$

(b) Static fluid: 4-velocity components

In the given coordinates G_{ab} is diagonal; for a perfect fluid (1) below

$$T^{ab} = (\rho + p/c^2)u^a u^b + g^{ab}p, \tag{1}$$

gives off-diagonal $T^{0i} \propto u^0 u^i$. Diagonality of G_{ab} forces $u^1 = u^2 = u^3 = 0$. The fluid is comoving with the static frame.

Normalisation $u^a u_a = -c^2$:

$$g_{00}(u^0)^2 = -c^2 \;\Rightarrow\; -e^{2\Phi}(u^0)^2 = -c^2.$$

Solve:

$$\boxed{u^0 = c\,e^{-\Phi}, \qquad u_0 = g_{00}u^0 = -c\,e^{\Phi}.}$$

(c) The 00 equation: enclosed-mass form of $\Lambda(r)$

$T_{00} = (\rho + p/c^2)u_0 u_0 + g_{00}p = (\rho c^2)\,e^{2\Phi}$. (Cross term: $(\rho + p/c^2)c^2 e^{2\Phi} - pe^{2\Phi} = \rho c^2 e^{2\Phi}$.) Field equation:

$$G_{00} = -\tfrac{8\pi G}{c^4}\rho c^2 e^{2\Phi}.$$

Substitute the given G_{00}:

$$-\frac{e^{2\Phi}}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\big[r(1 - e^{-2\Lambda})\big] = -\tfrac{8\pi G}{c^2}\rho\, e^{2\Phi}.$$

Cancel $e^{2\Phi}$ and rearrange:

$$\frac{\mathrm{d}}{\mathrm{d}r}\big[r(1 - e^{-2\Lambda})\big] = \tfrac{8\pi G}{c^2}\rho\, r^2.$$

Integrate from 0 to r with $r(1 - e^{-2\Lambda}) \to 0$ at the origin:

$$r(1 - e^{-2\Lambda}) = \tfrac{8\pi G}{c^2}\int_0^r \rho(r')r'^2\,\mathrm{d}r' = \frac{2GM(r)}{c^2}.$$

Solve for $e^{-2\Lambda}$:

$$\boxed{e^{-2\Lambda(r)} = 1 - \frac{2GM(r)}{c^2 r}.} \tag{8}$$

(d) The 11 equation: $\mathrm{d}\Phi/\mathrm{d}r$

$T_{11} = g_{11}p = e^{2\Lambda}p$. Field equation:

$$G_{11} = -\tfrac{8\pi G}{c^4}e^{2\Lambda}p.$$

Substitute given G_{11}:

$$\frac{e^{2\Lambda}}{r^2} - \frac{1}{r^2} - \frac{2\Phi'}{r} = -\tfrac{8\pi G}{c^4}e^{2\Lambda}p.$$

Multiply by $r^2 e^{-2\Lambda}$:

$$1 - e^{-2\Lambda} - 2r\,\Phi' e^{-2\Lambda} = -\tfrac{8\pi G}{c^4}r^2 p \cdot \text{(wait: re-multiply carefully)}.$$

Restart: rearrange the original equation algebraically. Multiply by r:

$$\frac{e^{2\Lambda}}{r} - \frac{1}{r} - 2\Phi' = -\tfrac{8\pi G}{c^4}r\,e^{2\Lambda}p.$$

Now multiply by $e^{-2\Lambda}$:

$$\frac{1}{r} - \frac{e^{-2\Lambda}}{r} - 2\Phi' e^{-2\Lambda} = -\tfrac{8\pi G}{c^4}r\,p.$$

Use (8): $1 - e^{-2\Lambda} = 2GM/(c^2 r)$. Hence $1/r - e^{-2\Lambda}/r = 2GM/(c^2r^2)$:

$$\frac{2GM(r)}{c^2r^2} - 2\Phi' e^{-2\Lambda} = -\tfrac{8\pi G}{c^4}r\,p.$$

Solve for Φ':

$$2\Phi' e^{-2\Lambda} = \frac{2GM(r)}{c^2r^2} + \frac{8\pi G p\,r}{c^4}.$$

Divide by $2e^{-2\Lambda} = 2(1 - 2GM/(c^2r))$:

$$\boxed{\frac{\mathrm{d}\Phi}{\mathrm{d}r} = \frac{GM(r) + 4\pi G p\,r^3/c^2}{c^2r^2\big(1 - 2GM(r)/(c^2r)\big)}.} \tag{9}$$

(e) TOV pressure equation

Energy–momentum conservation (given):

$$(\rho c^2 + p)\,\frac{\mathrm{d}\Phi}{\mathrm{d}r} = -\frac{\mathrm{d}p}{\mathrm{d}r}.$$

Substitute (9):

$$\boxed{\frac{\mathrm{d}p}{\mathrm{d}r} = -(\rho c^2 + p)\,\frac{GM(r) + 4\pi G p\,r^3/c^2}{c^2r^2\big(1 - 2GM(r)/(c^2r)\big)}.} \tag{10}$$

(f) Uniform-density star: central pressure

$\rho = \text{const}$, so $M(r) = \frac{4}{3}\pi\rho r^3$ for $r \le R$.

Convenient combination: $4\pi G\rho/c^2 = 3GM(R)/(c^2R^3) \cdot 1$ (consistent). Define

$$A \equiv \rho c^2, \qquad u(r) \equiv 2GM(r)/(c^2r) = \frac{8\pi G\rho r^2}{3c^2}.$$

Plug into (10): numerator inside the bracket is $GM(r) + 4\pi G p\, r^3/c^2 = G \cdot \frac{4}{3}\pi\rho r^3 + 4\pi G p r^3/c^2 = \frac{4\pi G r^3}{3c^2}(\rho c^2 + 3p)$.

$$\frac{\mathrm{d}p}{\mathrm{d}r} = -(\rho c^2 + p) \cdot \frac{4\pi G r^3(\rho c^2 + 3p)/(3c^2)}{c^2 r^2(1-u)} = -\frac{4\pi G r}{3c^4} \cdot \frac{(\rho c^2 + p)(\rho c^2 + 3p)}{1-u}.$$

Use $1 - u = 1 - 8\pi G\rho r^2/(3c^2)$. Separate variables:

$$\frac{\mathrm{d}p}{(\rho c^2 + p)(\rho c^2 + 3p)} = -\frac{4\pi G r/(3c^4)}{1 - 8\pi G\rho r^2/(3c^2)}\,\mathrm{d}r.$$

RHS is exact: let $w = 1 - 8\pi G\rho r^2/(3c^2)$, $\mathrm{d}w = -16\pi G\rho r/(3c^2)\,\mathrm{d}r$:

$$\text{RHS} = -\frac{4\pi G r}{3c^4} \cdot \frac{1}{w}\,\mathrm{d}r = \frac{1}{4\rho c^2} \cdot \frac{\mathrm{d}w}{w}.$$

Apply hint with $A = \rho c^2$, $x = p$:

$$\int \frac{\mathrm{d}p}{(A+3p)(A+p)} = \frac{3}{2A}\int \frac{\mathrm{d}p}{A+3p} - \frac{1}{2A}\int \frac{\mathrm{d}p}{A+p} = \frac{1}{2A}\left[\ln(A+3p) - \ln(A+p)\right].$$

Equate with LHS integrated:

$$\frac{1}{2A}\ln\frac{A+3p}{A+p} = \frac{1}{4A}\ln w + \text{const}.$$

Multiply by $2A$:

$$\ln\frac{A+3p}{A+p} = \tfrac{1}{2}\ln w + \text{const}.$$

Exponentiate:

$$\frac{A+3p}{A+p} = K\sqrt{w}.$$

Boundary at the surface $r = R$: $p(R) = 0$, $w(R) = 1 - 2GM/(c^2 R)$. Hence $K = 1/\sqrt{1 - 2GM/(c^2 R)}$. Total mass $M = \frac{4}{3}\pi\rho R^3$.

At the centre $r = 0$: $w(0) = 1$, so

$$\frac{A + 3p_0}{A + p_0} = \frac{1}{\sqrt{1 - 2GM/(c^2 R)}}.$$

Solve for p_0. Cross-multiply:

$$(A + 3p_0)\sqrt{1 - 2GM/(c^2 R)} = A + p_0.$$

Let $\sigma \equiv \sqrt{1 - 2GM/(c^2 R)}$:

$$A\sigma + 3p_0\sigma = A + p_0.$$

Group p_0 on one side:

$$p_0(3\sigma - 1) = A(1 - \sigma).$$

Solve and substitute $A = \rho c^2$:

$$\boxed{p_0 = \rho c^2\,\frac{1 - \sqrt{1 - 2GM/(c^2 R)}}{3\sqrt{1 - 2GM/(c^2 R)} - 1}.} \tag{11}$$

(g) Buchdahl bound

$p_0 \to \infty$ when the denominator of (11) vanishes:

$$3\sqrt{1 - 2GM/(c^2R)} = 1.$$

Square:

$$9(1 - 2GM/(c^2R)) = 1.$$

Solve:

$$\frac{2GM}{c^2R} = \frac{8}{9}.$$

Hence the upper bound on the mass at fixed R:

$$\boxed{M < \frac{4c^2R}{9G}.} \tag{12}$$

(h) Plausibility of $z = 3.1$

Surface gravitational redshift in the Schwarzschild exterior:

$$1 + z = \frac{1}{\sqrt{1 - 2GM/(c^2R)}}.$$

Solve for the compactness:

$$\frac{2GM}{c^2R} = 1 - \frac{1}{(1+z)^2}.$$

At $z = 3.1$, $(1 + z)^2 = 4.10^2 = 16.81$, so $1/(1 + z)^2 = 0.0595$:

$$\frac{2GM}{c^2R} = 0.940.$$

Compare with Buchdahl (12): $8/9 = 0.889$. Since $0.940 > 0.889$, the claimed redshift exceeds the maximum allowed by any static fluid configuration with $\rho \geq 0$ and ρ non-increasing outwards.

$$\boxed{\text{Claim implausible: } 2GM/(c^2R) = 0.94 > 8/9.}$$

Question 3 — FRW geometry and surface-brightness dimming

Problem. Realise the 3-sphere line element; state flat and hyperbolic counterparts; quote $4\pi r^2$-style surface areas. Justify the FRW form. Derive cosmological redshift. Compute angular diameter $\Delta\theta$ and the surface-brightness scaling, and discuss telescope diameter for high-z galaxies.

(a) The 3-sphere line element

Parametrise $w^2 + x^2 + y^2 + z^2 = A^2$ via

$$w = A\cos\chi, \quad x = A\sin\chi\sin\theta\cos\phi, \quad y = A\sin\chi\sin\theta\sin\phi, \quad z = A\sin\chi\cos\theta,$$

$\chi \in [0, \pi]$.

$$\mathrm{d}w^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 = A^2\left[\mathrm{d}\chi^2 + \sin^2\chi\,(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2)\right].$$

Change to $\bar{r} \equiv A \sin\chi$:

$$\mathrm{d}\bar{r} = A\cos\chi\,\mathrm{d}\chi, \qquad A^2\mathrm{d}\chi^2 = \frac{\mathrm{d}\bar{r}^2}{\cos^2\chi} = \frac{A^2\mathrm{d}\bar{r}^2}{A^2 - \bar{r}^2}.$$

Hence

$$\boxed{\mathrm{d}s^2_{(+1)} = \frac{A^2}{A^2 - \bar{r}^2}\,\mathrm{d}\bar{r}^2 + \bar{r}^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2).}$$

Flat ($k = 0$):

$$\mathrm{d}s^2_{(0)} = \mathrm{d}\bar{r}^2 + \bar{r}^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2).$$

Hyperbolic ($k = -1$): replace $\sin\chi \to \sinh\chi$; equivalently

$$\mathrm{d}s^2_{(-1)} = \frac{A^2}{A^2 + \bar{r}^2}\,\mathrm{d}\bar{r}^2 + \bar{r}^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2).$$

Surface area of $\bar{r} = \bar{R}$. The induced (θ, ϕ) metric is $\bar{R}^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2)$ in all three cases:

$$\boxed{\text{Area} = 4\pi\bar{R}^2 \quad \text{(flat, closed, open).}}$$

Geometry differs in the radial part, not in the area element.

(b) FRW from homogeneity and isotropy

Cosmological principle: the spatial 3-geometry is maximally symmetric (isotropic at every point). The only such 3-geometries are flat ($k = 0$), 3-sphere ($k = +1$) and 3-hyperboloid ($k = -1$). Allow the overall length scale to depend on time, $a(t)$: the spatial part of the line element is $a^2(t)\,\mathrm{d}\sigma_k^2$. Combining all three k cases by absorbing A into a and choosing r such that the angular part is a^2r^2:

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2(t)\left[\frac{\mathrm{d}r^2}{1 - kr^2} + r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2)\right],$$

with $k = 0, \pm 1$. The cosmic time t is the proper time of a comoving observer (lapse $-c^2$). Homogeneity guarantees a depends only on t; isotropy fixes the angular block as the round 2-sphere.

(c) Cosmological redshift

Radial null geodesics: $\mathrm{d}s^2 = 0$ on the equator with $\mathrm{d}\theta = \mathrm{d}\phi = 0$:

$$c\,\mathrm{d}t = \pm\,\frac{a(t)\,\mathrm{d}r}{\sqrt{1 - kr^2}}.$$

Integrate for one wavefront $tE \to t_0$:

$$\int_{t_E}^{t_0} \frac{c\,\mathrm{d}t}{a(t)} = \int_0^R \frac{\mathrm{d}r}{\sqrt{1 - kr^2}}.$$

The RHS is fixed by R (comoving). For a wavefront emitted one period later, $tE + \delta t_E \to t_0 + \delta t_0$, the comoving integral on the right is identical:

$$\frac{\delta t_E}{a(t_E)} = \frac{\delta t_0}{a(t_0)}.$$

Identify $\delta t = 1/\nu$:

$$\frac{\lambda_0}{\lambda_E} = \frac{a(t_0)}{a(t_E)}.$$

Definition $1+z = \lambda_0/\lambda_E$:

$$\boxed{1+z = \frac{a(t_0)}{a(t_E)}.} \tag{13}$$

(d) Angular diameter and surface brightness

Angular size. At emission time t_E, the proper transverse size of the disc is its diameter d. The proper transverse separation orthogonal to a radial null geodesic at $r = R$, $t = t_E$ is, from the FRW metric,

$$\mathrm{d}\ell_\perp = a(t_E)\, R\, \mathrm{d}\theta.$$

Set $\mathrm{d}\ell_\perp = d$:

$$\boxed{\Delta\theta = \frac{d}{a(t_E)R} = \frac{d(1+z)}{a(t_0)\, R}.} \tag{14}$$

Define the angular-diameter distance $d_A \equiv a(t_E)R$, so $\Delta\theta = d/d_A$. (At large z, $\Delta\theta$ can *increase* with z.)

Flux and surface brightness. Two factors of $(1+z)$ enter the flux: each photon redshifts in energy by $(1+z)$, and arrival rate at the detector is reduced by another $(1+z)$ (time dilation). The proper area of the wavefront sphere at present is $4\pi a^2(t_0)R^2$.

$$F = \frac{L}{4\pi a^2(t_0)R^2(1+z)^2}.$$

The luminosity distance is $d_L = a(t_0)R(1+z)$, with $F = L/(4\pi d_L^2)$.

Solid angle subtended:

$$\Delta\Omega = \pi(\Delta\theta/2)^2 = \frac{\pi d^2}{4\, a^2(t_E)R^2}.$$

Surface brightness:

$$\Sigma = \frac{F}{\Delta\Omega} = \frac{L}{4\pi a^2(t_0)R^2(1+z)^2} \cdot \frac{4\, a^2(t_E)R^2}{\pi d^2}.$$

Use $a(t_E) = a(t_0)/(1+z)$:

$$\Sigma = \frac{L}{\pi^2 d^2} \cdot \frac{1}{(1+z)^4}.$$

$$\boxed{\Sigma \propto \frac{1}{(1+z)^4} \text{ (Tolman dimming).}} \tag{15}$$

Implication for telescope diameter. A telescope of aperture D collects flux $\propto D^2$ and resolves a beam of solid angle $\Delta\Omega_{\mathrm{beam}} \sim (\lambda/D)^2$. The signal from a resolved galaxy is the surface brightness times the beam area times D^2:

$$S \propto \Sigma \cdot (\lambda/D)^2 \cdot D^2 = \Sigma\, \lambda^2.$$

The telescope-aperture factors cancel. Therefore for a galaxy that is *resolved* (extended source) increasing D does *not* improve the per-pixel brightness, and the $(1+z)^{-4}$ Tolman dimming cannot be defeated by aperture. (Larger D does help unresolved sources or pushes the resolution limit so that previously-unresolved high-z galaxies become resolved; once resolved, the limit is set by detector noise versus the dimmed surface brightness, and longer integration time, not larger aperture, is required.)

Question 4 — Thermal history: nucleosynthesis, decoupling, matter era

Problem. Compare nucleosynthesis and decoupling in the thermal history. Estimate n_γ today, n_γ/n_b, decoupling temperature, electron relativity, $a(t)$ in matter era, $T(t)$, age at decoupling.

(a) Nucleosynthesis vs decoupling

Big-Bang nucleosynthesis (BBN). Epoch $T \sim 0.1$–$1\,\mathrm{MeV}$, $t \sim 1$–$300\,\mathrm{s}$. Universe radiation-dominated. Weak interactions freeze out at $T_{\mathrm{wfo}} \sim 0.8\,\mathrm{MeV}$, fixing $n/p \approx 1/6$ (later $\to 1/7$ from neutron decay). Once the photon bath cools below the deuteron binding ($\sim 0.07\,\mathrm{MeV}$, photodissociation barrier; the actual nuclear chain proceeds at $T \sim 70\,\mathrm{keV}$ due to high η^{-1} entropy/baryon ratio — the "deuterium bottleneck"), nuclei build up: $D, {}^3\mathrm{He}, {}^4\mathrm{He}, {}^7\mathrm{Li}$. Primordial mass fractions: $Y_{^4\mathrm{He}} \approx 0.25$, $D/H \sim 10^{-5}$.

Decoupling (recombination). Epoch $T \sim 0.3\,\mathrm{eV}$, $t \sim 3.8 \times 10^5\,\mathrm{yr}$. Photons cease Thomson-scattering off free electrons because the electrons combine with protons to form neutral hydrogen: $e^- + p \to H + \gamma$. The CMB photons free-stream from the surface of last scattering. The temperature is much below the H ionisation energy $13.6\,\mathrm{eV}$ because the high photon-to-baryon ratio $\eta^{-1} \sim 10^9$ keeps a tail of ionising photons until $T \ll 13.6\,\mathrm{eV}$.

Common features. Both occur in the radiation-dominated era ($H \propto T^2$, $t \propto T^{-2}$); both are out-of-equilibrium freeze-out events governed by the competition between an interaction rate Γ and the Hubble rate H ($\Gamma \sim H$ defines the freeze-out temperature); both leave fossil relics (light elements, CMB).

Differences. BBN fixes the chemical composition of baryons; decoupling fixes the photon mean free path. BBN is much hotter and earlier, governed by strong/weak nuclear physics; decoupling is governed by atomic physics. Photons are tightly coupled in both, but only at decoupling do they free-stream.

(b) CMB spectrum and present temperature

The CMB is a near-perfect blackbody at

$$\boxed{T_0 = 2.725\,\mathrm{K}.}$$

Spectral form: Planck distribution

$$u_\nu\,\mathrm{d}\nu = \frac{8\pi h\nu^3/c^3}{e^{h\nu/k_BT} - 1}\,\mathrm{d}\nu.$$

Typical photon energy: $E_\gamma \sim 2.7 k_B T$ (mean for blackbody). Compute:

$$k_B T_0 = 1.381 \times 10^{-23} \times 2.725 = 3.76 \times 10^{-23}\,\mathrm{J}.$$

$$E_\gamma \approx 2.7 \times 3.76 \times 10^{-23} = 1.02 \times 10^{-22}\,\mathrm{J}.$$

Photon number density (energy density / energy per photon):

$$n_\gamma \sim \frac{u_\gamma}{E_\gamma} = \frac{4 \times 10^{-14}}{1.02 \times 10^{-22}}.$$

Divide:

$$n_\gamma \approx 3.9 \times 10^8\,\mathrm{m}^{-3} \approx \boxed{4 \times 10^8\,\mathrm{m}^{-3}.}$$

(Exact Planck integration gives $4.1 \times 10^8\,\mathrm{m}^{-3}$.)

(c) Photon-to-baryon ratio and decoupling temperature

Critical density: $\rho_c = 3H_0^2/(8\pi G) \approx 9 \times 10^{-27}\,\mathrm{kg\,m^{-3}}$.

Baryon density (using $\Omega_b \approx 0.05$):

$$n_b = \Omega_b \rho_c / m_p = 0.05 \times 9 \times 10^{-27}/1.67 \times 10^{-27} = 0.27\,\mathrm{m^{-3}}.$$

Photon-to-baryon ratio:

$$\eta^{-1} = n_\gamma / n_b = 4 \times 10^8 / 0.27 \approx \boxed{1.5 \times 10^9.}$$

Decoupling temperature. Saha-style estimate: recombination occurs when the fraction of photons in the high-energy tail with $E > E_{\mathrm{ion}} = 13.6\,\mathrm{eV}$ becomes of order $\eta = n_b/n_\gamma \sim 10^{-9}$.

$$\frac{n_\gamma(E > E_{\mathrm{ion}})}{n_\gamma} \sim e^{-E_{\mathrm{ion}}/k_B T_{\mathrm{dec}}} \sim \eta.$$

Solve:

$$E_{\mathrm{ion}}/(k_B T_{\mathrm{dec}}) \sim \ln(1/\eta) = \ln\bigl(1.5 \times 10^9\bigr) = 21.$$

$$k_B T_{\mathrm{dec}} \sim 13.6/21 = 0.65\,\mathrm{eV}.$$

$$T_{\mathrm{dec}} \sim 0.65/k_B[\mathrm{eV/K}] = 0.65/8.617 \times 10^{-5} \approx \boxed{7.5 \times 10^3\,\mathrm{K}.}$$

The full Saha calculation gives $T_{\mathrm{dec}} \approx 3000\,\mathrm{K}$; the order-of-magnitude estimate above is high because we used a crude Boltzmann tail and neglected the ionisation-equilibrium prefactor (Saha) and that recombination is more efficient than the simple tail estimate suggests. *Approximations:* (i) Boltzmann tail in place of full Saha; (ii) neglect of two-photon decay channel; (iii) sharp instantaneous recombination.

Are the electrons relativistic? Compare $k_B T_{\mathrm{dec}}$ (or $\sim 0.3\,\mathrm{eV}$ for the accurate value) with $m_e c^2 = 511\,\mathrm{keV}$:

$$k_B T_{\mathrm{dec}}/m_e c^2 \sim 1\,\mathrm{eV}/5.11 \times 10^5\,\mathrm{eV} \sim 2 \times 10^{-6} \ll 1.$$

$$\boxed{\text{Electrons strongly non-relativistic at decoupling.}}$$

(d) Matter era: $a(t)$ and $T(t)$; age at decoupling

Friedmann (flat, matter-dominated, $\rho \propto a^{-3}$):

$$\left(\frac{\dot a}{a}\right)^2 = \frac{8\pi G}{3}\rho = \frac{8\pi G}{3}\rho_0\, a^{-3}.$$

Take square root:

$$\dot a = C\, a^{-1/2}, \qquad C \equiv \sqrt{8\pi G \rho_0/3}.$$

Separate variables:

$$a^{1/2}\,\mathrm{d}a = C\,\mathrm{d}t.$$

Integrate:

$$\tfrac{2}{3} a^{3/2} = Ct.$$

$$\boxed{a(t) \propto t^{2/3} \text{ (EdS, matter era).}}$$

Temperature. Use (13): $T \propto 1/a$ (CMB redshift), so

$$T(t) \propto t^{-2/3}.$$

Age scaling: $t \propto T^{-3/2}$.

Age at decoupling. Use today's age $t_0 \sim 1.4 \times 10^{10}\,\mathrm{yr}$ and $T_0 = 2.725\,\mathrm{K}$, $T_{\rm dec} \approx 3000\,\mathrm{K}$ (the textbook value):

$$t_{\rm dec} = t_0 \left(\frac{T_0}{T_{\rm dec}} \right)^{3/2} = 1.4 \times 10^{10} \times \left(\frac{2.725}{3000} \right)^{3/2}.$$

Ratio:

$$2.725/3000 = 9.08 \times 10^{-4}.$$

Power 3/2:

$$(9.08 \times 10^{-4})^{3/2} = (9.08 \times 10^{-4}) \times \sqrt{9.08 \times 10^{-4}} = 9.08 \times 10^{-4} \times 3.01 \times 10^{-2} = 2.74 \times 10^{-5}.$$

Multiply:

$$t_{\rm dec} = 1.4 \times 10^{10} \times 2.74 \times 10^{-5} = 3.8 \times 10^{5}\,\mathrm{yr}.$$

$$\boxed{t_{\rm dec} \sim 4 \times 10^{5}\,\mathrm{yr}.}$$

(Matter–radiation equality occurs earlier, at $z_{\rm eq} \sim 3400$, so the strict $t \propto T^{-3/2}$ scaling is good for $z \lesssim 1100$, and the estimate is reliable.)